

◇ AUTONOMOUS WEATHER INTELLIGENCE PLATFORM

WlaaS

Weather Intelligence as a Service

Transforming raw atmospheric forecasts into autonomous, physical mitigation strategies across Asia.

DEVELOPMENT PLATFORM: BUILT WITH QODER APP

REGIONAL SCOPE: 29 ASIAN SOVEREIGN TERRITORIES

◇ DIAGNOSTICS

FIG. 02

The Operational Disconnect

01

THE INDUSTRIAL REALITY

- ◇ **Observation vs. Action:** Current platforms predict numbers — *"it will be 42°C"* — and leave humans to guess the operational impact.
- ◇ **Systemic Vulnerability:** Grid failures, agricultural losses, and logistics disruptions occur rapidly, often faster than manual response cycles.
- ◇ **The Core Issue:** A massive disconnect exists between atmospheric physics and immediate operational decision-making.

DECISION GAP: 4.5h TO 12.0h HUMAN DELAY PREVENTS PROACTIVE FIELD DEFENSE

02

CURRENT INDUSTRY CAPABILITY GAP



CRITICAL FAILURE: 95% SATELLITE DATA NEVER AUTOMATES IMMEDIATE MITIGATION

◇ RESPONSE ARCHITECTURE

The WlaaS Solution

FIG. 03

01

THE BRIDGE

An end-to-end platform that fuses a deterministic physics engine (**FastAPI**) with a collaborative AI swarm (**n8n**) — converting raw atmospheric signal into coordinated field action.

DATA INGEST → PHYSICS ENGINE → 11 AI AGENTS → FIELD MITIGATION

The bridge eliminates speculative LLM reasoning. Physical anomaly bounds strictly constrain all agent deliberations before instructions render to operators.

02

ACTION OVER OBSERVATION

WlaaS does not just tell you what will happen — it applies environmental equations and autonomously orders immediate mitigation actions.

EXAMPLE 01 • TETENS SATURATION VAPOR PRESSURE:

$$e_s(T) = 0.61078 \cdot \exp((17.27 \cdot T) / (T + 237.3)) \text{ kPa}$$

EXAMPLE 02 • TRIGGERING MITIGATION VIA VAPOR PRESSURE DEFICIT:

$$VPD = e_s - e_a$$

SURVIVABILITY BOUNDARY: $T_{wb} \geq 35.0^{\circ}\text{C} \rightarrow$ AUTONOMOUS HEAT-SHOCK WORK BAN

◇ STACK OVERVIEW

FIG. 04

System Architecture

01

DATA INGESTION

Pulling real-world, live climate and disaster telemetry via the Open-Meteo API, NASA FIRMS thermal hotspots, and RainViewer radar mesh.

Feeds Ingested:

- Open-Meteo 72h Nowcast & Ensemble
- NASA FIRMS Satellite Thermal Scan
- RainViewer Doppler Radar Tiles
- GDACS & GDELT Real-Time OSINT

02

PHYSICS ENGINE (FASTAPI)

Real-time mathematical computation of thermal anomalies, irrigation efficiency, and 24-hour power grid load predictions with zero latency.

Solvers Executed:

- Tetens Thermodynamic VPD Solver
- Stull Wet-Bulb Globe (WBGT)
- Synthetic Ledger (Evap, MW, Fuel)
- Synchronous Grid Predictor

03

COGNITIVE LAYER (N8N SWARM)

11 specialized AI agents across two consolidated workflows collaborate sequentially to generate a unified crisis plan in < **2,800 tokens**.

Swarm Division:

- 7 Agents in WlaaS Assistant Graph
- 4 Agents in CrisisLens Graph
- Zero-Token Deterministic Gating
- Compact State Vector (<350 bytes)

DEVELOPMENT & VALIDATION:

BUILT AND TESTED USING QODER APP

◇ TOKEN OPTIMIZATION

FIG. 05

Token-Efficient Architecture

01 OPERATING PRINCIPLE: Rules, APIs and memory handle predictable work first. The LLM is activated only when interpretation, correlation or action generation is genuinely required.

02

DETERMINISTIC CORE

Intent Router: Classifies greeting, FAQ, direct metric before model call.

Zero-Token Regex: Rules answer predictable requests with 0 LLM.

Conditional Run: Executes only required specialists.

Fast vs. Deep Path: Direct answers stay lightweight.

03

CONTEXT COMPRESSION

Compact State: Reduces payloads to <350 bytes needed for reasoning.

API Normalization: Mixed outputs into one small evidence vector.

Relevance Filter: Cuts distant events before model.

Limited Context: Agents get only task fields.

04

MEMORY & OUTPUT

Session Memory: Stores region, assessment for multi-turn reuse.

Cached Routes: Severity and evidence reuse prior state.

Strict JSON: Predefined fields prevent verbose outputs.

No-Invention Rule: Returns fallback rather than speculating.

05

MODEL & EFFICIENCY

Fireworks + MiniMax: Fast inference & reliable instruction following.

Token Benchmark:

- Naive Swarms: 16k–22k
- LangChain ReAct: 9k–14k
- WlaaS: <2,800 (-82%)

Response Latency: <200ms

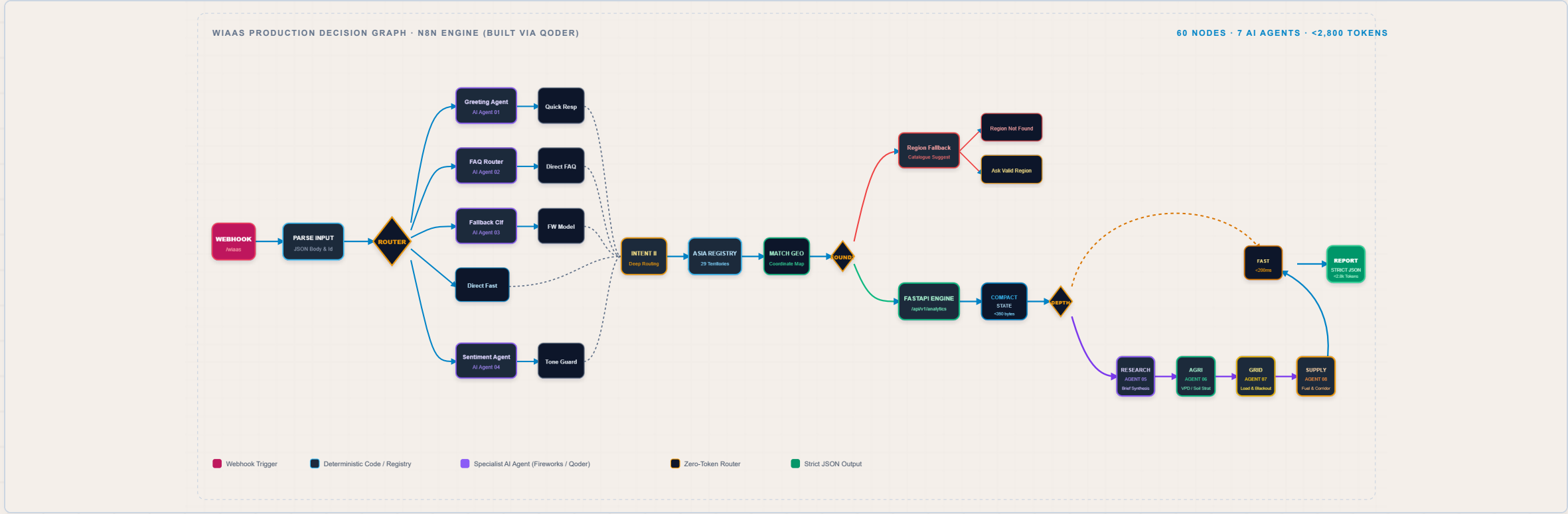
USER REQUEST → DETERMINISTIC ROUTE → COMPACT EVIDENCE → FIREWORKS AI (IF NEEDED) → STRICT JSON RESPONSE

WORKFLOW MAP

WiaaS Assistant Workflow

60 PRODUCTION NODES · 7 SPECIALIZED AI AGENTS

Simplified logical view of the production architecture. The full WiaaS Assistant contains 60 n8n nodes; repeated validation, parsing, fallback, memory, API, agent and response steps are grouped into representative blocks below.



DATA / CODE: Parses input, calls APIs, normalizes payloads

ROUTER: Chooses intent, region, fast/deep execution

MEMORY: Reuses session context & prior state

AI AGENT: Interprets evidence & generates sector actions

OUTPUT: Builds unified JSON

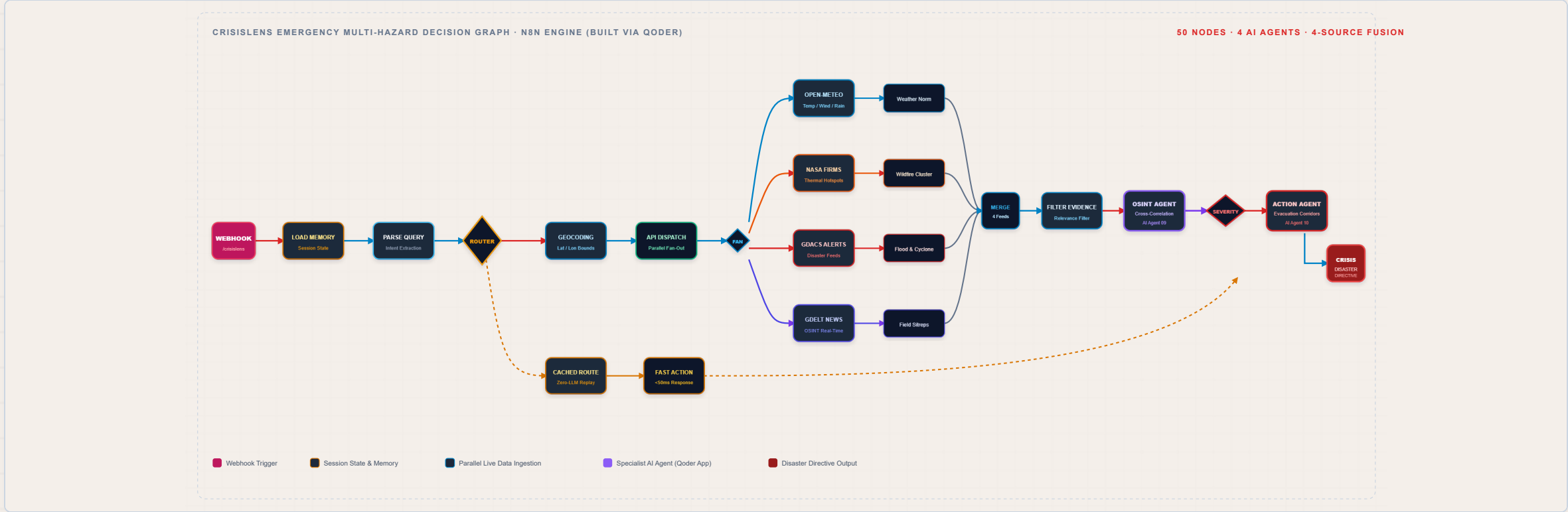
WORKFLOW MAP

FIG. 07

CrisisLens Workflow

50 PRODUCTION NODES · 4 AI AGENTS · 4-SOURCE FUSION

Simplified logical view of the production threat detection architecture. The full CrisisLens workflow contains 50 n8n nodes; repeated session, geocoding, parallel evidence, filtering, confidence, action and response steps are grouped below.



INPUT / MEMORY: Loads message & previous assessment

ROUTING / GEO: Resolves intent & coordinates

EVIDENCE: 4 Parallel live feeds (Weather, FIRMS, GDACS, GDELT)

AI REASONING: Correlates evidence & confidence

OUTPUT: Evacuation protocol

◇ LIVE SIMULATION & ASIAN RESILIENCE MATRIX

FIG. 08

Jaipur Heatwave & Asian Matrix

01

ACTIVE CRISIS ANALYSIS

Deterministic prediction of a **15% grid blackout risk** at 7:00 PM (19:00 LT), driven by compounding thermal anomalies and non-linear transformer load derating.

02

AUTONOMOUS EXECUTION

Automatic triggering of protocols for immediate human work restrictions under Stull WBGT 35°C criteria, and preventative fuel load staging across primary distribution feeders.

03

GLOBAL SCALABILITY

Integrated region registry capable of mapping, resolving, and simulating risk across **420+ cities** and 29 sovereign Asian territories instantly.

BLACKOUT RISK WINDOW
Compounding thermal peak threshold

19:00 LT

SIMULATED REGION COVERAGE
Granular sub-district resolution

420+ Cities

Multan / S. Punjab: Night irrigation shift (21:00-04:00) saves 40% water evaporation; transformer derating -18%.

Karachi / Sindh: Maritime wet-bulb defense ($T_{wb}=34.2^{\circ}\text{C}$), cooling centers, Lyari/Malir floodgates.

Tokyo / Kanto: Typhoon wind shear structural greenhouse bracing & BTM battery islanding.

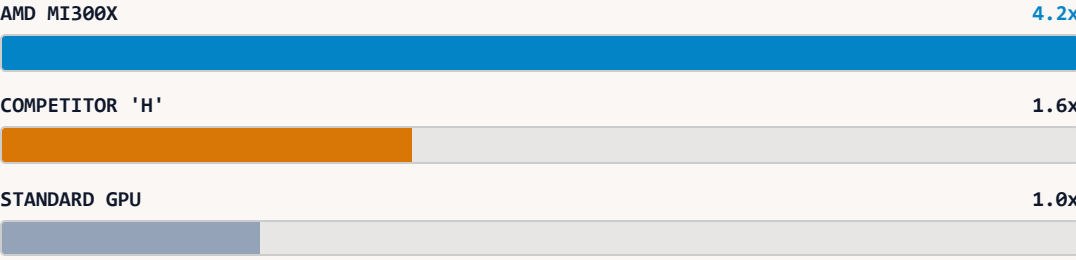
◇ HARDWARE PARTNER

FIG. 09

Powered by AMD Instinct™ MI300X

01

RELATIVE SIMULATION THROUGHPUT



Throughput advantage driven primarily by the 192GB HBM3 VRAM capacity, enabling zero-latency co-location of physics and cognitive models.

02

BREAKING THE COMPUTE BOTTLENECK

Traditional processors saturate under real-time geospatial physics and multi-agent reasoning. The AMD CDNA™ 3 architecture provides massively parallel processing.

03

CO-LOCATING PHYSICS & COGNITION

Running local swarms alongside deep-physics models demands immense memory. The MI300X delivers **192 GB of HBM3 VRAM** and **5.3 TB/s** bandwidth.

04

ZERO-LATENCY SYNERGY

The entire GNN pipeline and agentic models load onto the exact same card, eliminating PCIe bottlenecks.

◇ ENGINEERING DISCIPLINE

FIG. 10

The WIaaS Advantage

01

ABSOLUTE RELIABILITY

Zero AI hallucinations. Every agent response is mathematically constrained and strictly guided by real metrics from the backend physics engine.

02

OPTIMIZATION AT SCALE

A complex, multi-agent operational architecture is compressed into less than **2,800 tokens** per full run, achieving extreme efficiency.

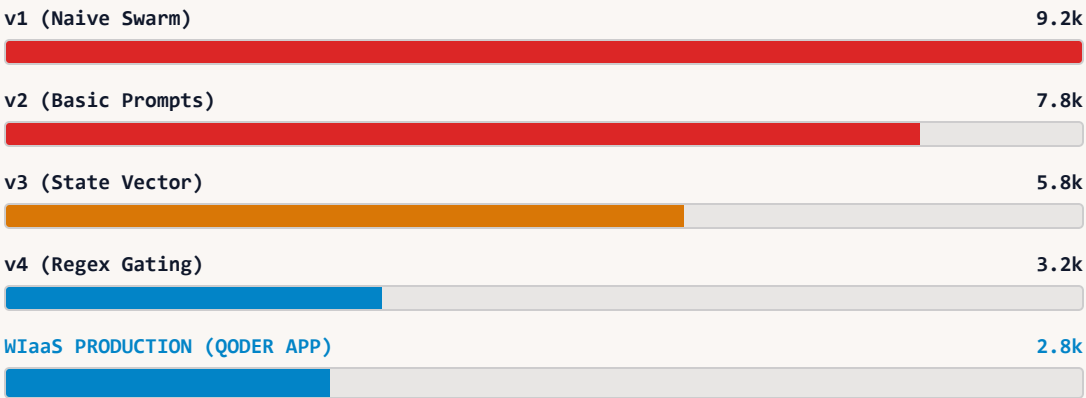
03

HARDWARE-ACCELERATED INTERACTION

Sustained 60 FPS MapLibre GL Thermographic map, 1-second dynamic streaming canvas oscilloscope, and neural bilingual speech (English & Urdu).

03

TOKEN USAGE — PER FULL SIMULATION RUN



82% TOKEN COMPRESSION DELIVERS SUB-SECOND LATENCY AT FRACTIONAL COST

◇ COMMERCIAL MODEL

FIG. 11

Business Model & Economy

01

MONETIZATION STRATEGY

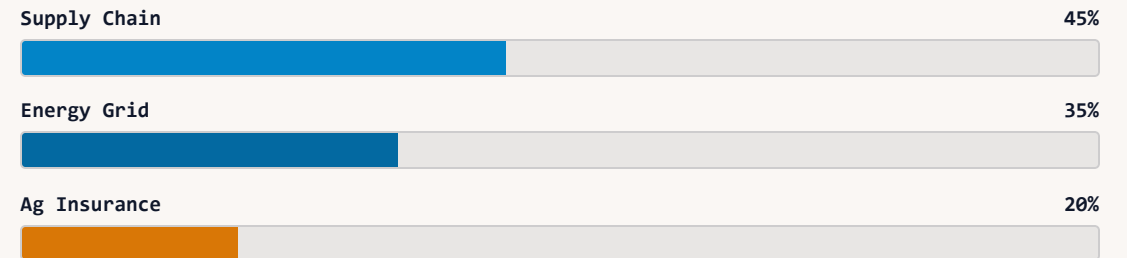
B2B SaaS & API. Tiered subscription model for enterprise and grid operators, alongside a pay-per-simulation API for AgTech platforms.

"Mitigating physical climate risk is no longer a luxury; it is a baseline requirement for scaling businesses."

TARGET SECTORS:

- ◇ **Supply Chain:** Protecting high-value perishables and cold-chain routing.
- ◇ **Energy Grid:** Proactive load-shedding intelligence and transformer protection.
- ◇ **Ag Insurance:** Parametric micro-insurance payouts based on verified physics thresholds.

02

REVENUE COMPOSITION

TOTAL REGIONAL CLIMATE RISK ADDRESSABLE MARKET: \$18.4B BY 2030

◇ FORWARD TRAJECTORY

FIG. 12

Roadmap & Vision

STAGE 01 · VALIDATION

FOUNDATION

Core engine development, API integration (Open-Meteo), Tetens thermodynamic solvers, and the Jaipur Heatwave live simulation.

STATUS: COMPLETED & VERIFIED

STAGE 02 · DATA EXPANSION

TELEMETRY SCALE

Immediate integration of global crisis telemetry streams, including NASA FIRMS and GDACS. 11-Agent swarm across 2 n8n graphs, MapLibre GL 60 FPS GIS, mobile sheets.

BUILT & SHIPPED WITH QODER APP

STAGE 03 · ENTERPRISE SCALE

FULL DEPLOYMENT

Large-scale commercial deployment, automated SCADA/grid load shedding, Satellite SAR flood tracking, optimized for AMD Instinct™ accelerated environments.

COMMERCIAL ROLLOUT

AUTONOMOUS RESILIENCE:

WIaaS · ENGINEERED WITH QODER APP